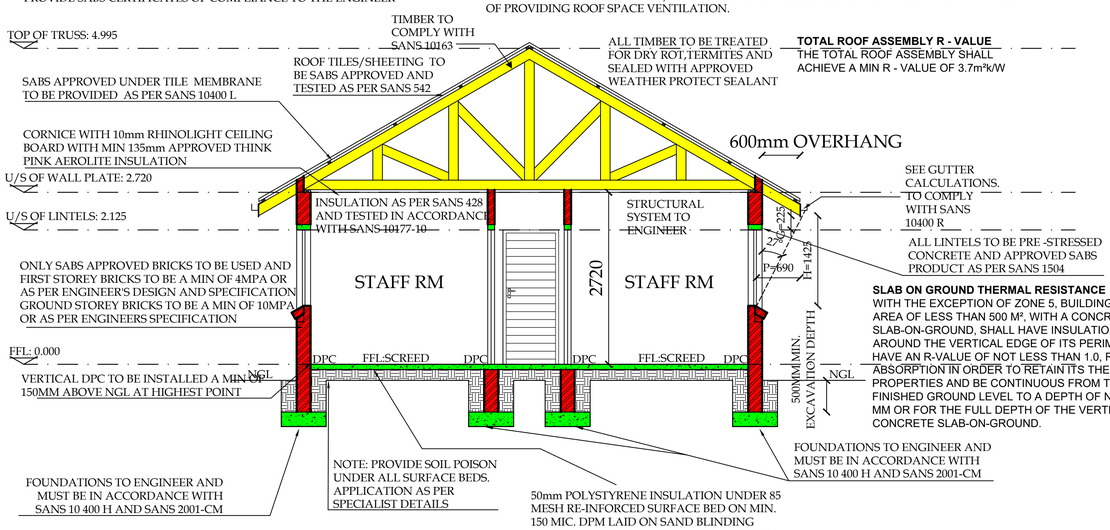
SHADING
WHERE SHADING IS USED, THE BUILDING SHALL HAVE A PERMANENT FEATURE SUCH AS A VERANDA, BALCONY, FIXED CANOPY, EAVES OR SHADING HOOD WHICH EXTENDS HORIZONTALLY ON BOTH SIDES OF THE GLAZING FOR THE SAME PROJECTION DISTANCE OR PROVIDES THE EQUIVALENT SHADING WITH A REVEAL OR OTHER SHADING ELEMENT OR HAVE AN EXTERNAL SHADING DEVICE, SUCH AS A SHUTTER, BLIND, VERTICAL OR HORIZONTAL BUILDING SCREEN WITH BLADES, BATTENS OR SLATS, WHICH IS CAPABLE OF RESTRICTING AT LEAST 80 % OF SUMMER SOLAR RADIATION AND IF ADJUSTABLE, IS READILY OPERATED EITHER MANUALLY, MECHANICALLY OR ELECTRONICALLY BY THE BUILDING OCCUPANTS.

THE CONTRACTOR SHALL TAKE ALL NECESSARY MEASUREMENTS AND DIMENSIONS ON SITE PRIOR TO MANUFACTURING AND INSTALLATION. THE CONTRACTOR SHALL PROVIDE SABS CERTIFICATES OF COMPLIANCE TO THE ENGINEER

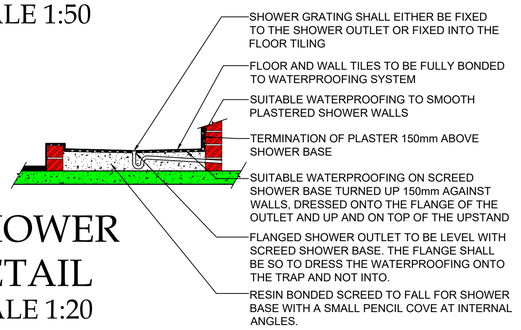
ROOF GAPS AND VENTILATION
CARE SHOULD BE TAKEN TO ENSURE THAT THE ROOF VENTILATION OPENINGS DO NOT ALLOW RAIN PENETRATION. GAPS BETWEEN ROOF TILES WITH SARKING (OR REFLECTIVE INSULATION AT RAFTER LEVEL) AND METAL SHEET ROOFING ARE NOT ACCEPTABLE METHODS OF PROVIDING ROOF SPACE VENTILATION.



TOTAL ROOF ASSEMBLY R - VALUE
THE TOTAL ROOF ASSEMBLY SHALL ACHIEVE A MIN R - VALUE OF 3.7m²k/W

SLAB ON GROUND THERMAL RESISTANCE
WITH THE EXCEPTION OF ZONE 5, BUILDINGS WITH A FLOOR AREA OF LESS THAN 500 M², WITH A CONCRETE SLAB-ON-GROUND, SHALL HAVE INSULATION INSTALLED AROUND THE VERTICAL EDGE OF ITS PERIMETER WHICH SHALL HAVE AN R-VALUE OF NOT LESS THAN 1.0, RESIST WATER ABSORPTION IN ORDER TO RETAIN ITS THERMAL INSULATION PROPERTIES AND BE CONTINUOUS FROM THE ADJACENT FINISHED GROUND LEVEL TO A DEPTH OF NOT LESS THAN 300 MM OR FOR THE FULL DEPTH OF THE VERTICAL EDGE OF THE CONCRETE SLAB-ON-GROUND.

SECTION A-A
SCALE 1:50



SHOWER DETAIL
SCALE 1:20

TABLE 5.1.9

MINIMUM NUMBER OF COURSES	MAXIMUM ROOF SPAN		
	NO ROOF	LIGHT ROOF	HEAVY ROOF
85MM COURSE HEIGHT:	NOMINAL THICKNESS ≤ 140mm		
4	3.0	2.0	1.5
5	3.0	2.5	2.0
6	3.0	3.0	2.5
9	3.0	3.0	3.0
85MM COURSE HEIGHT:	NOMINAL THICKNESS < 190mm		
4	3.0	2.0	2.0
5	3.5	2.5	2.5
6	3.5	3.5	3.0
9	3.5	3.5	3.5
100MM COURSE HEIGHT:	NOMINAL THICKNESS < 140mm		
3	3.0	2.0	1.5
4	3.0	2.5	2.0
5	3.0	3.0	2.5
8	3.0	3.0	2.5
100MM COURSE HEIGHT:	NOMINAL THICKNESS < 190mm		
3	2.5	2.0	2.0
4	3.0	2.5	2.0
5	3.5	3.5	3.0
8	3.5	3.5	3.5

LINTELS THAT SUPPORT ROOF TRUSSES TO BE BUILT INTO BRICKWORK:
1. LINTELS WITH A SPAN OF 1.5m OR LESS = 150MM
2. SPAN OF 1.5M - 2.5M = 250MM
3. SPAN GREATER THAN 2.5M = 350MM
LINTELS TO BE SUPPORTED ADEQUATELY FOR NOT LESS THAN 7 DAYS AFTER COMPLETION

PRE-STRESSED CONCRETE LINTELS :
PRECAST CONCRETE LINTELS SHALL COMPLY WITH THE RELEVANT REQUIREMENTS OF SABS 1504 AND BE BUILT INTO WALLS COMPOSITELY WITH MASONRY. SUCH LINTELS SHALL BE CONSTRUCTED IN ACCORDANCE WITH TABLE 5.8 AND CONTAIN NOT LESS THAN THE NUMBER OF COURSES SET OUT IN TABLE 5.2
SECONDARY REINFORCEMENT IN ACCORDANCE WITH TABLE 5.1.5 SHALL BE PROVIDED IN UPPERMOST BED JOINTS. LINTELS SHALL BE SET IN CLASS ii MORTAR AND HAVE A MINIMUM BEARING OF:

* LINTELS SUPPORTING MASONRY ONLY =150mm
* LINTELS SUPPORTING ROOF TRUSSES
  SPAN <1.5m =150mm
  SPAN1.5-2.5m=250mm
  SPAN >2.5m =350mm

LINTELS SHALL BE CLEANED AND WETTED PRIOR TO RECEIVING MORTAR

ROOF CALCULATIONS:

CLIMATE ZONE 1 DIRECTION OF AIR FLOW -UP

MIN REQUIRED R VALUE =3.7M²K/W

THERMAL CONDUCTIVITY =K VALUE = W/MK

$$\frac{\text{THERMAL CONDUCTIVITY}}{\text{THICKNESS OF MATERIAL}} = \text{U VALUE}$$

$$\text{U=1/R} \qquad \text{R=1/U}$$

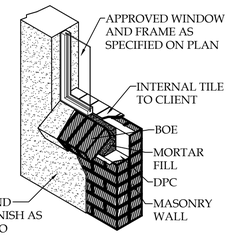
$$\frac{0.040\text{W/M.K}}{1\text{M}} = 0.040\text{W/M}^2\text{.K}$$

$$\frac{1}{0.040\text{W/M}^2\text{K}} = 25\text{M}^2\text{.K/W}$$

TILED ROOF WITH PLASTERED CEILING BOARD =0.40M²K/W
3.7-0.40=3.30

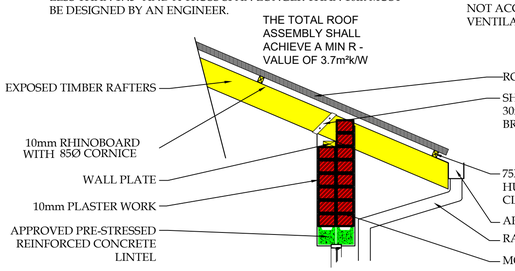
$$\frac{3.30}{25}$$

ADD 135MM THINK PINK INSULATION



BOE WINDOW CILL DETAIL
SCALE 1:20

THE HOLDING DOWN FOR SHEETED ROOFS WITH A PITCH OF LESS THAN 17.5° AND A TRUSS SPAN LONGER THAN 10m MUST BE DESIGNED BY AN ENGINEER.



SHEETED ROOF DETAIL
SCALE 1:20

ROOF GAPS AND VENTILATION
CARE SHOULD BE TAKEN TO ENSURE THAT THE ROOF VENTILATION OPENINGS DO NOT ALLOW RAIN PENETRATION. GAPS BETWEEN ROOF TILES WITH SARKING (OR REFLECTIVE INSULATION AT RAFTER LEVEL) AND METAL SHEET ROOFING ARE NOT ACCEPTABLE METHODS OF PROVIDING ROOF SPACE VENTILATION.

SPECIFICATIONS

1. GENERAL
1.1 THIS DRAWING IS NOT TO BE SCALED, USE FIGURED DIMENSIONS ONLY. ALL DIMENSIONS AND HEIGHTS TO BE CHECKED AND VERIFIED BEFORE ANY WORK COMMENCES ON SITE. ANY DISCREPANCIES SHALL BE REPORTED TO THE ARCHITECT IMMEDIATELY. ALL LEVELS, HEIGHTS OF PLINTHS, DEPTH OF EXCAVATIONS AND NUMBER OF STEPS TO BE FINALLY CHECKED BY CONTRACTOR ON SITE.
1.2 THE SITE TO BE TREATED IN ACCORDANCE WITH S.A.B.S. CODE OF PRACTICE NO.0124-1977 WITH 'SHELLDRITE' TERMITEPRUFE SOIL POISONER.
1.3 TOP OF FOUNDATIONS TO BE A MINIMUM OF 300mm BELOW NATURAL GROUND LEVEL.
1.4 BACKFILL TO FOUNDATIONS.
1.5 TOP OF 75mm CONCRETE SURFACE BED TO BE MINIMUM OF 150mm ABOVE FINISHED GROUND LEVEL
1.6 75mm THICK CONCRETE SURFACE BED TO BE ON 'GUNPLAS' USB GREEN DAMP PROOF MEMBRANE ON SAND BINDING ON WELL COMPACTED FILL.
1.7 'GUNDLE' BRICKGRIP S.A.B.S. EMBOSSED D.P.C 375 MIC. UNDER ALL WALLS, WINDOW CILLS AND AT CHANGES IN FLOOR LEVELS.
1.8 STORMWATER SHALL BE REMOVED FROM DWELLING, YARD AND SITE.
1.9 ELECTRICAL INSTALLATION SHALL BE 'HEINEMANN' EARTH LEAKAGE SYSTEM IN DISTRIBUTION BOARD.
1.10 ALL BUILDING WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES BUILDING BY-LAWS AND REGULATIONS.

2. FOUNDATIONS
2.1 ALL FOUNDATIONS, RETAINING WALLS, STRUCTURAL CONCRETE WORK AND SUB SOIL STORMWATER DRAINAGE TO CIVIL ENG'S SPECIFICATIONS.
2.2 ALL SOIL COMPACTION TO CIVIL ENG'S SPECIFICATIONS.
2.3 ALL FOUNDATIONS TO BE REINFORCED AS REQUIRED BY ENGINEER.

3. FLOORS
3.1 MIN. 75mm CONCRETE SURFACE BED.
3.2 25mm CEMENT SCREED TO ALL FLOORS.
3.3 SKIRTINGS TO BE PROVIDED.

4. WALLS
4.1 220mm BRICK WALLS WITH GUNDLE BRICKGRIP DPC.
4.2 BRICKFORCE EVERY 4 COURSES AND EVERY COURSE FOR 4 COURSES OVER OPENINGS
4.3 ALL BRICKWORK MIN. 7 Mpa CLASS II.
4.4 WINDOWS BUILT IN WITH GUNDLE BRICKGRIP DPC.
4.5 ALL PAINTWORK TO COMPLY WITH MANUFACTURER'S SPECIFICATIONS WITH NECESSARY UNDERCOATS.
4.6 EXTERNAL-TO PLAN
INTERNAL-PLASTER AND PAINT.

ALL LEVELS, DIMENSIONS, DEPTHS OF EXCAVATIONS, HEIGHTS OF PLINTHS, NUMBER OF STEPS AND SIZES OF FOUNDATIONS TO BE PLACED AT A HEIGHT OF EXISTING STRUCTURE

5. ROOF A
5.1 CROMADEK SHEETING @ 30° PITCH- DARK DOLPHIN ON 38X50mm S.A. PINE PURLINS @ 1200mm MAX C/C. ON 152X38mm S.A.PINE, GRADE 6, PREFABRICATED TRUSSES @ 1200 C/C ON S.A. PINE 114X38mm WALLPLATES. TRUSSES TO BE TIED DOWN WITH HOOP IRONS THREADED THROUGH FOUR COURSES OF BRICKWORK. PROVIDE SABS APPROVED PLASTIC UNDERLAY AND SISALATION INSULATION.
5.2 MIN.170mm CONCRETE SLAB TO ENG'S DETAIL.
5.3 MIN. 55mm VEMICULITE INSULATING SCREED TO 1:60 FALL.
5.4 WATERPROOFING BY SPECIALIST TO MANUFACTURERS SPEC.
5.5 150Ø P.V.C RAINWATER DOWNPIPES FROM FULLBORE OUTLETS.
5.6 SKYLIGHTS TO MANUFACTURERS SPEC.
5.7 GUTTERS : MARLEY STREAMLINE SQUARE PROFILE RAINWATER GUTTERS + DOWNPIPES.
5.7 3mmSKIM COAT RHINOLITE ON PLASTERBOARD CEILING FIXED TO 38 X 38mm S.A. PINE BRANDERING AT 450 C/C FIXED TO TIE BEAMS.

6. WINDOWS AND DOORS
6.1a ANY PANE OF GLASS WHICH IS TO BE INSTALLED WITHOUT A SUPPORT FRAME SHALL BE IN ACCORDANCE WITH SABS 0137
6.1b THICKNESS OF PANES IN RELATION TO THEIR AREA SHALL BE IN ACCORDANCE WITH SABS 0137
6.2 EXTERNAL SLIDING DOORS : STANDARD AS PER WINDOWS TO SCHEDULE.
6.3 INTERNAL DOORS : STANDARD HARDWOOD FRAMES TO SCHEDULE
6.4 WINDOWS : ALUMINUM TOP-HUNG
6.5 SLIDING DOORS :STANDARD ALUMINIUM SLIDING DOORS TO SCHEDULE.

7. DRAINAGE NOTES
7.1 100Ø uPVC. SEWER PIPE DRAIN WITH A MIN. FALL OF 1:60.
7.2 100Ø OVP. AT HEAD OF DRAIN PIPE AS SHOWN.
7.3 RODDING EYES AT HEAD OF DRAIN, AT ALL CHANGES OF DIRECTION AND AT MAX. OF 25000 INTERVALS.
7.4 INSPECTION EYES AT ALL JUNCTIONS OF DRAIN, AND TO HAVE MARKED COVERS AT GROUND LEVEL.
7.5 DRAIN PIPE UNDER BUILDING TO BE PROTECTED AGAINST LOAD.
7.6 ALL WASTE PIPE TO HAVE 65mm RE-SEAL TRAPS, ALL WASTE PIPES TO BE ACCESSIBLE OVER ENTIRE LENGTH FOR CLEANING AND REPAIRS.
7.7 ALL WASTE PIPES UNDER FLOOR SLAB TO BE SLEEVED.
7.8 ALL SOIL FITTINGS WITH VERTICAL DISCHARGE GREATER THAN 1220 TO HAVE ANTISYPHON VENTPIPES.
7.9 RAINWATER DOWNPIPES TO DISCHARGE A MIN. OF 2440 FROM ANY OPEN GULLY
7.10 ALL DRAINAGE WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES DRAINAGE BY-LAWS AND REGULATIONS.

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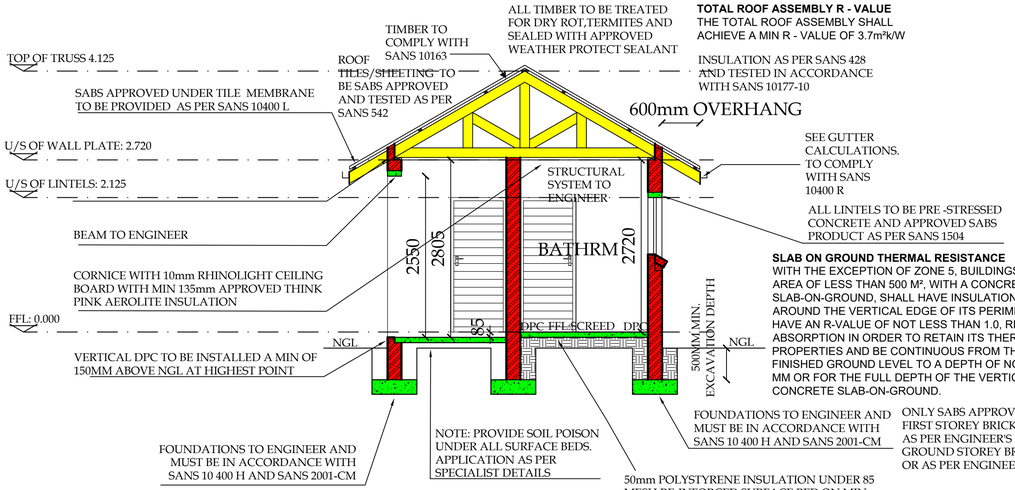


SHADING
WHERE SHADING IS USED, THE BUILDING SHALL HAVE A PERMANENT FEATURE SUCH AS A VERANDA, BALCONY, FIXED CANOPY, EAVES OR SHADING HOOD WHICH EXTENDS HORIZONTALLY ON BOTH SIDES OF THE GLAZING FOR THE SAME PROJECTION DISTANCE OR PROVIDES THE EQUIVALENT SHADING WITH A REVEAL OR OTHER SHADING ELEMENT OR HAVE AN EXTERNAL SHADING DEVICE, SUCH AS A SHUTTER, BLIND, VERTICAL OR HORIZONTAL BUILDING SCREEN WITH BLADES, BATTENS OR SLATS, WHICH IS CAPABLE OF RESTRICTING AT LEAST 80 % OF SUMMER SOLAR RADIATION AND IF ADJUSTABLE, IS READILY OPERATED EITHER MANUALLY, MECHANICALLY OR ELECTRONICALLY BY THE BUILDING OCCUPANTS.

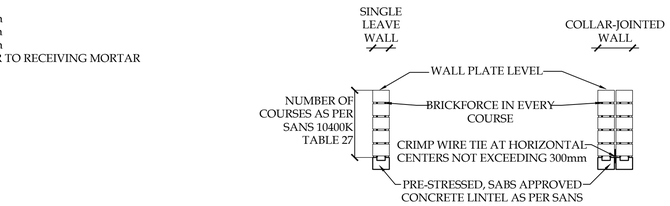
TOTAL ROOF ASSEMBLY R - VALUE
THE TOTAL ROOF ASSEMBLY SHALL ACHIEVE A MIN R - VALUE OF 3.7m²k/W

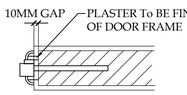
INSULATION AS PER SANS 428 AND TESTED IN ACCORDANCE WITH SANS 10177-10

SLAB ON GROUND THERMAL RESISTANCE
WITH THE EXCEPTION OF ZONE 5, BUILDINGS WITH A FLOOR AREA OF LESS THAN 500 M², WITH A CONCRETE SLAB-ON-GROUND, SHALL HAVE INSULATION INSTALLED AROUND THE VERTICAL EDGE OF ITS PERIMETER WHICH SHALL HAVE AN R-VALUE OF NOT LESS THAN 1.0, RESIST WATER ABSORPTION IN ORDER TO RETAIN ITS THERMAL INSULATION PROPERTIES AND BE CONTINUOUS FROM THE ADJACENT FINISHED GROUND LEVEL TO A DEPTH OF NOT LESS THAN 300 MM OR FOR THE FULL DEPTH OF THE VERTICAL EDGE OF THE CONCRETE SLAB-ON-GROUND.

ONLY SABS APPROVED BRICKS TO BE USED AND FIRST STOREY BRICKS TO BE A MIN OF 4MPA OR AS PER ENGINEER'S DESIGN AND SPECIFICATION GROUND STOREY BRICKS TO BE A MIN OF 10MPA OR AS PER ENGINEERS SPECIFICATION

SECTION B-B
SCALE 1:50





SOLID TIMBER FRAME DETAIL
SCALE N.T.S

TYPICAL LINTEL DETAIL
SCALE 1:20

BRICKFORCE SHOULD SPAN MIN. 300mm OVER AT SIDES AND 5.6mm RODS MIN 600mm OVER AT SIDES OF OPENING.
BRICKFORCE AT CENTERS THAT DO NOT EXCEED 200mm

ALL DIMENSIONS TO BE MEASURED OUT AND PEGGED ON SITE BEFORE COMMENCING WORK

ALL REINFORCED CONCRETE SLABS, FOUNDATIONS, COLUMN DETAILS, BEAMS, STAIRS AND RETAINING WALLS TO BE BUILT STRICTLY IN ACCORDANCE WITH ENGINEERS DETAIL AND UNDER HIS/HER SUPERVISION

BUILDING MUST BE IN ACCORDANCE WITH THE APPROVED BUILDING PLAN. NO CHANGES MAY BE MADE TO SUCH PLANS WITHOUT THE EXPRESSED CONSENT OF THE AUTHOR OF SUCH DOCUMENT

FLOORS,WALLS,GEYSER PIPES AND ROOFS TO BE INSULATED IN ACCORDANCE WITH SANS 10400

ARCHITECTURAL DRAWINGS TO BE READ IN CONJUNCTION WITH ENGINEERING DRAWINGS

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ACTS OF PARLIAMENT

All Contractors shall ensure that, before any work is put in hand, they comply with all the necessary Acts of Parliament of the Republic of South Africa.

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PROPOSED STAFF ROOMS
ON PTN 465
OF THE FARM KROMDRAAI 560KQ
CLIENT:

WORKING DRAWINGS

TECHNOLOGISTS:	W JUNG-KRUGER
ARCHITECT:	

DATE:OCT'18	DWG NO: H/KD
DRAWN: WJK	SHEET : 3 OF 6